HARIKRISHNAN SRIKRISHNAN

Phone: +1 602-677-7372 | Email: hsrikris@asu.edu | Portfolio: https://hksk-create.github.io/Portfolio/

ACADEMIC QUALIFICATION

Master of Science (Computer Science), Arizona State University (ASU), Tempe, AZ, United States
Expected Graduation: May 2027 | GPA: 3.78

Bachelor of Engineering (Computer Science and Engineering), Sri Sivasubramaniya Nadar College of Engineering (SSN), Chennai, India; CGPA: 8.2, **August 2018 – May 2022**

WORK EXPERIENCE

Software Engineer, Toyota Connected India (TCIN), Chennai, India

Duration: July 2022 – September 2025

- Responsible for the safety and security of the vehicle occupants, including alerting in case of accidents and thefts, offering roadside assistance to vehicles, collision detection and so much more.
- Utilized Salesforce CRM to track and manage customer and vehicle data across agent-facing workflows, and led a UX overhaul of 4 case screens — redesigning key components for visual clarity, reorganizing information hierarchy, decluttering redundant fields, and fixing critical broken components — enabling agents to more efficiently resolve 100–200 monthly vehicle safety incidents.

Project 1: System Design Challenge | Team Size: 4

- **Role:** Analyzing the overall architecture, keeping an eye on vulnerabilities
- Secured the 3rd place for designing the architecture from scratch for the European electric market, where domestic charging stations could be used due to the scarcity of regular electric stations, with efficient and robust system components

Project 2: Hackathon | Team Size: 6

- **Role**: Front-end development using React, React-NodeJS integration
- Displayed heatmaps on a Google Maps interface to alert drivers about accident-prone zones along his/her route; Displayed the places along the route that would need a permit to enter.

PERSONAL PROJECTS

Resume – Job Description Matcher (Full-Stack Web Application)

Tech: FastAPI (Python-based web framework), React, SQLite, SentenceTransformers (Python ML library), Recharts

- Built a full-stack Resume–JD matching platform using FastAPI (REST APIs) and React with SQLite for persistent storage.
- Implemented hybrid matching logic combining semantic similarity (transformer embeddings) and dynamic keyword scoring with alias-based skill normalization.
- Designed a feature that estimates how adding specific missing skills would improve the resume–JD match score.
- Developed an interactive analytics dashboard featuring skill demand trends, impact ranking, and role competitiveness insights using Recharts (React charting library).
- Structured backend services with modular endpoints for analysis, dashboard metrics, role inference etc. improving maintainability and scalability.

ACADEMIC PROJECTS

Ensuring end-to-end exactly-once semantics during stream processing using Apache Kafka and Spark
Duration: February 2022 – March 2022
Team Size: 3

Role: Integrated Apache Kafka and Spark, implemented the project in HDFS-based clusters with a name node and three data nodes; Used the Cassandra Clusters as a database to prevent loss in case of node failures.
Objective: Designed and benchmarked two fault-tolerance approaches — unique ID generation and Kafka offset checkpointing — to guarantee exactly-once semantics across a Kafka → Spark → Cassandra streaming pipeline, validated against up to 5 million records.
Technology/Tools Used: Apache Kafka, Spark

Health Hive
Duration: February 2020 – March 2020
Team Size: 3
Role: Backend and Frontend development
Objective: To store and handle patient-related data like Patient ID, Patient name, Temperature, Disease, Doctor ID, and Doctor name; insert, delete, or update details; Filter patients using the above fields and display them when required.
Technology/Tools Used: Backend - Java, Frontend – Swing

Book Vault
Duration: August 2019 – September 2019
Team Size: 2
Role: Backend and Frontend development
Objective: Log in with a unique username and password, add new books, remove existing books, search for a particular book based on a unique ISDN number and impose fines if the book is returned after the date of return.
Technology/Tools Used: Backend - Python, Frontend - Tkinter

CERTIFICATION COURSES

- Master Machine Learning for Data Science and AI – Codebasics, Ongoing
- React – Udemy, Ongoing
- Docker and Kubernetes – Udemy, Ongoing
- Git & GitHub, June 2025
- Reinforcement Learning – NPTEL (National Programme on Technology Enhanced Learning), September 2021
- Data Analytics with Python – MyCaptain, December 2020

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, HTML5, CSS
Frameworks & Tools: React, Node.js, FastAPI, Docker, Kubernetes, Apache Kafka, Spark, Salesforce, Git

VOLUNTEERING ACTIVITIES

Member, National Service Scheme (NSS), SSN College, 2018 - 2021

- Met and interacted with students in a distant village through an annual event to impart knowledge and explore different lifestyles/viewpoints, July 2019
- Engaged in multiple campus cleaning activities as part of a “clean and green” initiative, June 2019.

EXTRA-CURRICULAR ACTIVITIES

- Conducted a technical symposium, a quiz competition organized in periodic intervals by the Computer Science Department, SSN College, July 2021